When Correlation Isn’t Enough

Using Causal Methods for Discriminatory Bias Detection

Joyce S. Padela, Pavan Raj Ravi, Jianwu Wang
Department of Information Systems
University of Maryland, Baltimore County
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Problem Statement

- Fairness audits mostly rely on correlation metrics
  - Statistical parity, equalized odds, disparate impact ratio (four-fifths rule) [4, 5, 6, 9]
  - They answer “Is there a disparity?” — not “Where does it come from?”
- They cannot separate direct discrimination from proxy-mediated effects
  - Removing a sensitive feature does not remove a proxy (e.g., ZIP code, prior convictions)
- COMPAS: among defendants who did not reoffend, 44.9% of African-American vs. 23.5% of Caucasian defendants were labeled medium/high risk [3, 8]
  - Does race act on the score directly, or through priors shaped by upstream policing?
- Without a mechanism, interventions are guesswork

Research Question and Contributions

- **To what extent can causal discovery algorithms identify structural patterns in data that produce biased outcomes?**
- **A five-step causal pipeline for bias detection**
  - Six algorithms (PC, FCI, GES, GRaSP, ICA-LiNGAM, DirectLiNGAM) [14 -16, 18-21] benchmarked against planted ground truth
  - Sensitivity analysis over 480 runs (2,880 fits) charts which method works at which sample size and effect size
- **Real-world COMPAS application**
  - LiNGAM variants recover a direct Race → Score edge; DirectLiNGAM matches the adjusted direct effect within 0.0088 decile points
- **A fairness-audit failure mode for BIC-based permutation search (GRaSP [21])**
  - Risk scores built from their own inputs ($R^2 = 0.45$) look exogenous, producing impossible edges into immutable attributes

Validating Fairness: *A Causal Inference Pipeline for Discriminatory Bias Assessment*

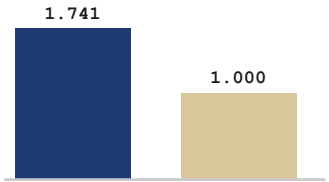
Correlation reveals **that** disparity exists. Causal discovery reveals **where** it comes from — and **how** to intervene. Demonstrated on ProPublica COMPAS (n=5,278).

STEP 01 — DETECT

Establish the Correlation Baseline

Compute correlation-based fairness metrics (Disparate Impact Ratio, selection-rate gap, rate ratios) to confirm a disparity is present.

SELECTION-RATE DISPARITY (COMPAS)



COMPAS CORRELATION METRICS

Selection-Rate DIR	1.741
Recidivism Rate Ratio	1.338
4/5 Rule (inverted)	0.575 FAILS

Correlation detects that bias exists — but offers no mechanistic explanation of its source.

STEP 02 — DISCOVER

Select a Causal Discovery Algorithm

Six algorithms across four families were benchmarked on synthetic data with planted ground-truth bias (β = −0.15).

ALGORITHM FAMILIES TESTED

CONSTRAINT-BASED
PC · FCI
≤15% detection at n ≤ 10,000, all β. Improves only at n=50,000 and erratically. Returns equivalence classes—orientation often undetermined.

SCORE-BASED (BIC)
GES
35-75% rising to 75-100% at larger β and n.

PERMUTATION
GRaSP
Peaks at 70% (n=5,000), declines to 45% at n=50,000. Highly initialization-sensitive. Misorients scores built from their own predictors (high-R^2 exogeneity misclassification, Step 3).

FUNCTIONAL CAUSAL — PREFERRED
ICA-LiNGAM · DirectLiNGAM
≥85% detection at n=1,000; recovers oriented edges and β magnitudes.

WHY DIRECTLINGAM FOR REAL-WORLD ASSESSMENT

Exploits non-Gaussianity · guaranteed convergence · no ICA hyperparameters · robust to binary & heavy-tailed variables typical of high-stakes administrative data.

Algorithm choice is not methodologically neutral — it determines which forms of discriminatory bias remain visible.

STEP 03 — ISOLATE

Isolate Direct vs. Proxy Discrimination

Map the recovered DAG to separate two structurally distinct discrimination pathways — each demanding a different intervention.

▸ DIRECT PATHWAY

Race → Score

Discriminatory bias embedded *inside* the scoring rules themselves. **Intervention: revise the algorithm's internal logic.**

▸ PROXY PATHWAY

Race → Priors → Score

Structural inequities (e.g., racially disparate policing) inflate a "neutral" risk signal. **Intervention: upstream policy reform — removing Race alone will not eliminate this path.**

WATCH FOR: HIGH R^2 EXOGENEITY MISCLASSIFICATION

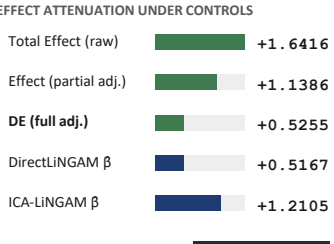
BIC-ordering heuristics (e.g., GRaSP) can misclassify the high R^2 outcomes as exogenous — producing implausible arrows into immutable attributes like Race or Age. Avoid purely BIC-driven permutation search when scores are highly collinear with their inputs.

STEP 04 — QUANTIFY

Quantify the Causal Effect

Apply covariate adjustment to estimate the total effect and the adjusted direct effect (DE) — blocking criminal-history mediators {Age, Charge Degree, Juv Fel, Juv Misd, Priors}.

EFFECT ATTENUATION UNDER CONTROLS



CONVERGENT CROSS-VALIDATION

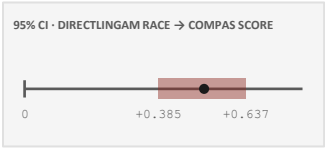
Adjusted DE (regression) and DirectLiNGAM (non-Gaussianity) — differing identification strategies on the same data — agree within **0.0088** decile-score points, isolating discriminatory bias from legitimate risk signals.

A residual gap survives full criminal-history controls — consistent with direct discriminatory bias.

STEP 05 — VALIDATE

Validate with Bootstrapping

1,000 non-parametric bootstrap resamples (fixed seed) generate percentile-based 95% confidence intervals to test statistical stability.



ALL THREE CIS EXCLUDE ZERO

DirectLiNGAM 95% CI	[+0.385, +0.637]
DE (full) 95% CI	[+0.395, +0.663]
ICA-LiNGAM 95% CI	[+1.076, +1.345]
E-value (sensitivity)	≈ 1.65

Reject the null of zero bias.
Convergent point estimates, overlapping CIs excluding zero, and an E-value comparable to the strongest known criminogenic risk factors make a confounding explanation unlikely.

STEP 01 **Detect** — *Does disparity exist?* → STEP 02 **Discover** — *Which algorithm can see it?* → STEP 03 **Isolate** — *Direct or via a proxy?* → STEP 04 **Quantify** — *How large, after controls?* → STEP 05 **Validate** — *Stable & not by chance?*

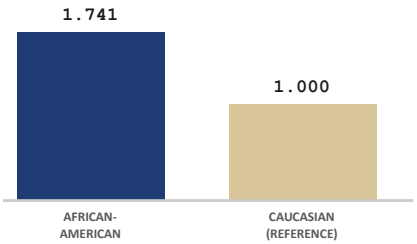
Step 1 - Detect: Establish the Correlation Baseline

STEP 01 — DETECT

Establish the Correlation Baseline

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COMPAS metric (n = 5,278)	African-Am.	Caucasian	Ratio / gap
Defendants	3,175	2,103	—
Mean decile score (1–10)	5.2769	3.6353	gap 1.6416
High-risk rate, Score ≥ 5	**0.5761**	**0.3310**	**1.7406**
Four-fifths rule (inverted)			**0.5745 < 0.80**
Two-year recidivism rate	0.5231	0.3909	1.3384

Correlation confirms a disparity exists — but cannot say whether Race acts on the Score directly or through prior convictions.

ProPublica filters on compas-scores-two-years (n = 6,172), restricted to African-American and Caucasian defendants. The DIR is defined for binary selection rates, so we use the Score ≥ 5 high-risk rate.

CI from 1,000 unconstrained bootstrap refits (point estimate +0.510).

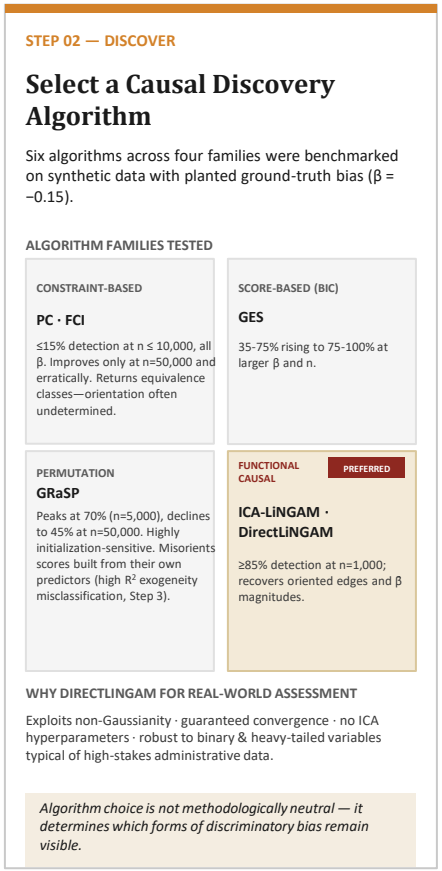
Methodology

Synthetic loan-approval data (ground truth)

- Paired datasets, n = 5,000: A biased (β = −0.15), B unbiased (β = 0)
- We use causal-learn [22] and DoWhy [23] packages for the causal discovery and inference algorithms
- Planted direct edge Race → Loan; proxy path Race → ZIP → CreditSc → Loan; latent SES confounder
- β = −0.15 sits within documented gaps in hiring, healthcare, and lending [24, 25, 26]
- Metric: Structural Hamming Distance (SHD) and detection of the oriented Race → Loan edge
- Sensitivity: β ∈ {0–0.25} × n ∈ {1k, 5k, 10k, 50k}, 20 seeds → 480 runs, 2,880 fits

Empirical data: COMPAS [3, 8]

- ProPublica compas-scores-two-years, standard filters → n = 5,278
- Nine variables on natural scales (decile score 1–10)
- DirectLiNGAM with background knowledge: Race, Sex, Age are exogenous; recidivism a sink
- Baseline: disparate impact ratio (four-fifths rule)
- Effects: total and covariate-adjusted direct effect (OLS); 1,000 bootstrap resamples; E-value [27]

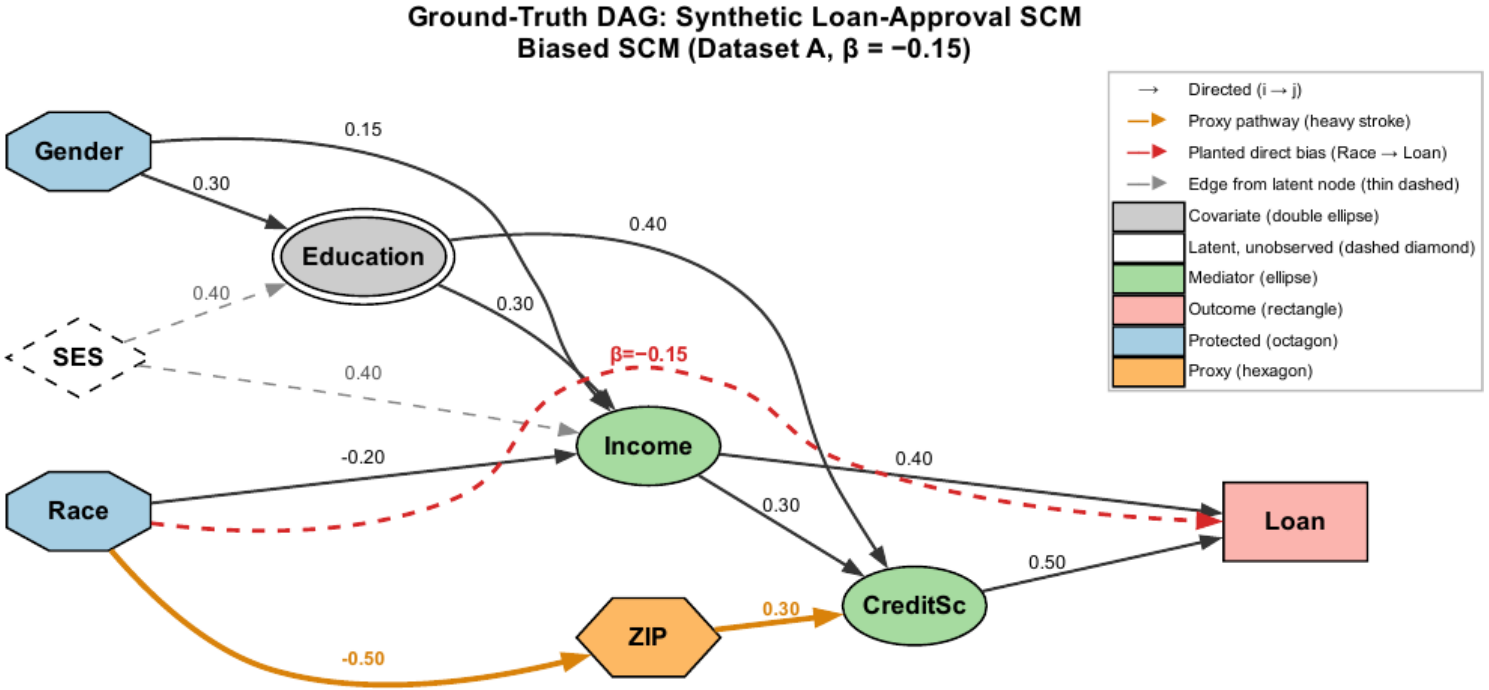


Six Causal Discovery Algorithms

Only the functional causal family (LiNGAM) commits to unique edge directions and estimates β. The planted bias β = −0.15 lowers loan approval for minority applicants.

Algorithm	Family	What it does	Detects latent confounder	Unique edge directions	Identifies β
PC (Peter-Clark) [19]	Constraint	Tests statistical independence	No	Partial	No
FCI (Fast Causal Inference)	Constraint +	Detects hidden confounders	Yes	Partial	No
GES (Greedy Equivalence Search) [20]	Score (BIC)	Searches Markov equivalence classes for the best score	No	Partial	No
GRaSP (Greedy Relaxation of the Sparsest Permutation) [21]	Permutation	Greedy search over variable orderings; relaxes GES	No	Partial	No
ICA-LiNGAM [15]	Func. causal	Linear non-Gaussian acyclic model via ICA	No	Full	**Yes**
DirectLiNGAM [16] **Preferred**	Func. causal	Direct entropy-based LiNGAM; robust to binary and heavy-tailed variables	No	Full	**Yes**

Synthetic Dataset Ground Truth



Biased dataset A. Dashed red = planted direct bias Race → Loan (β = −0.15). Orange = proxy path Race → ZIP → CreditSc → Loan (present in A and B). SES = latent confounder of Education and Income.

Synthetic Results: Recovering the Planted Bias

Algorithm	SHD A	SHD B	Edge (A)	Edge (B)	Latent conf.	β (A)
PC	8	7	No	No	No	–
FCI	9	8	No	No	Yes	–
GES	13	3	No	No	No	–
GRaSP	8	7	No	No	No	–
ICA-LiNGAM	**1**	**1**	**Yes**	**No**	**No**	**−0.1074**
DirectLiNGAM	**1**	**1**	**Yes**	**No**	**No**	**−0.1074**

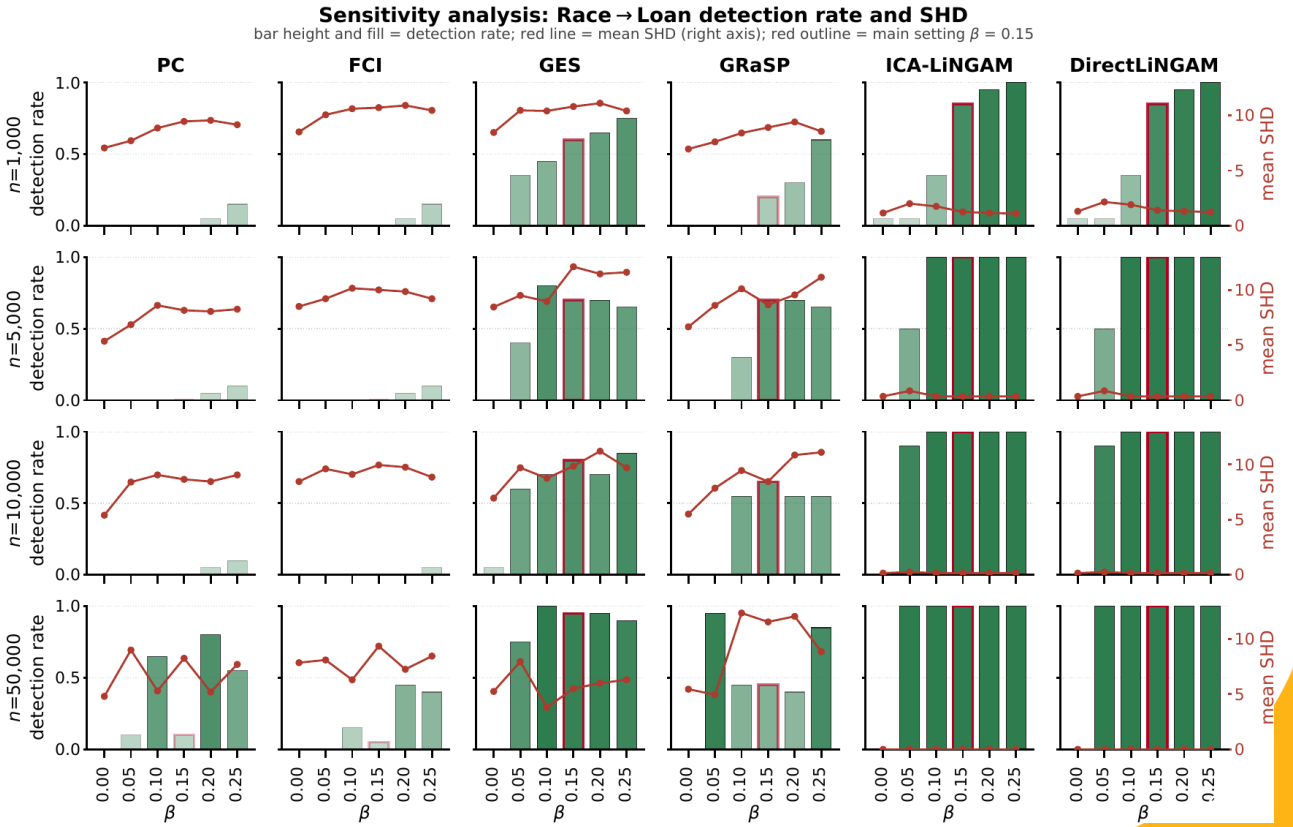
n = 5,000. A = biased, B = unbiased (β for B = 0.000). Edge = oriented Race → Loan detected. SHD = edits to reach the ground-truth graph; lower is better.

- Both LiNGAM variants recover the planted Race → Loan edge in A and correctly omit it in B (SHD = 1)
- β̂ = −0.1074 (planted −0.15); the adjusted direct effect −0.1086 agrees within 0.0012
- FCI flags the latent SES confounder (undirected Education–Income edge)
- PC, FCI, GES, and GRaSP miss the oriented edge
- GRaSP is initialization-sensitive: 14 of 20 seeds detect the edge; the pinned run does not

Step 2 - continued

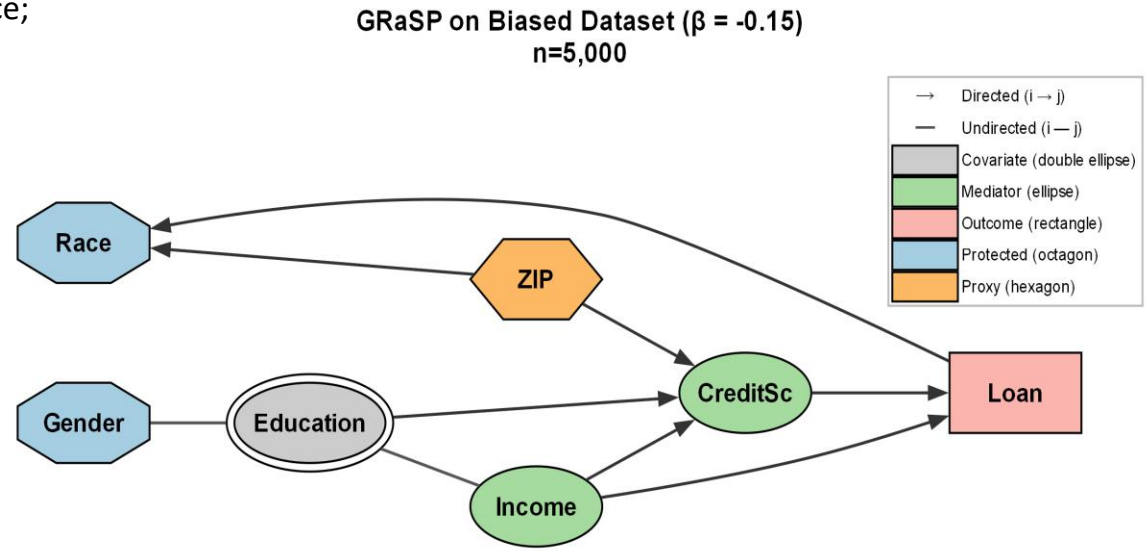
Sensitivity Analysis: Who Detects the Bias, and When?

- PC, FCI: ≤ 15% detection at n ≤ 10,000, every β
- GES: improves slowly with sample size
- GRaSP: peaks at 70% (n = 5,000), then falls to 45% at n = 50,000
- LiNGAM: ≥ 85% at n = 1,000; ≥ 95% at n = 5,000 for β ≥ 0.10
- Practical floor for audits: β ≈ 0.05
- **Algorithm choice is not neutral**



Audit Failure Mode: GRaSP on Risk Scores

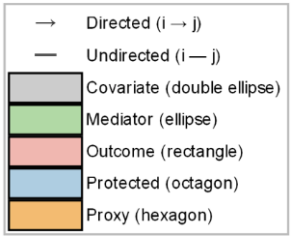
- GRaSP orders variables by residual variance; the lowest is treated as most exogenous
- A risk score built from its own inputs has artificially low residual variance
  - → placed first, with arrows into immutable attributes
- More data gives a more confident wrong answer (SHD 8.65 → 11.55)
- Red flag: any arrow into Race, Sex, or Age
- COMPAS Score $R^2 = 0.45$; BIC prefers the impossible graph by ≈ 70 points (see supplemental material).

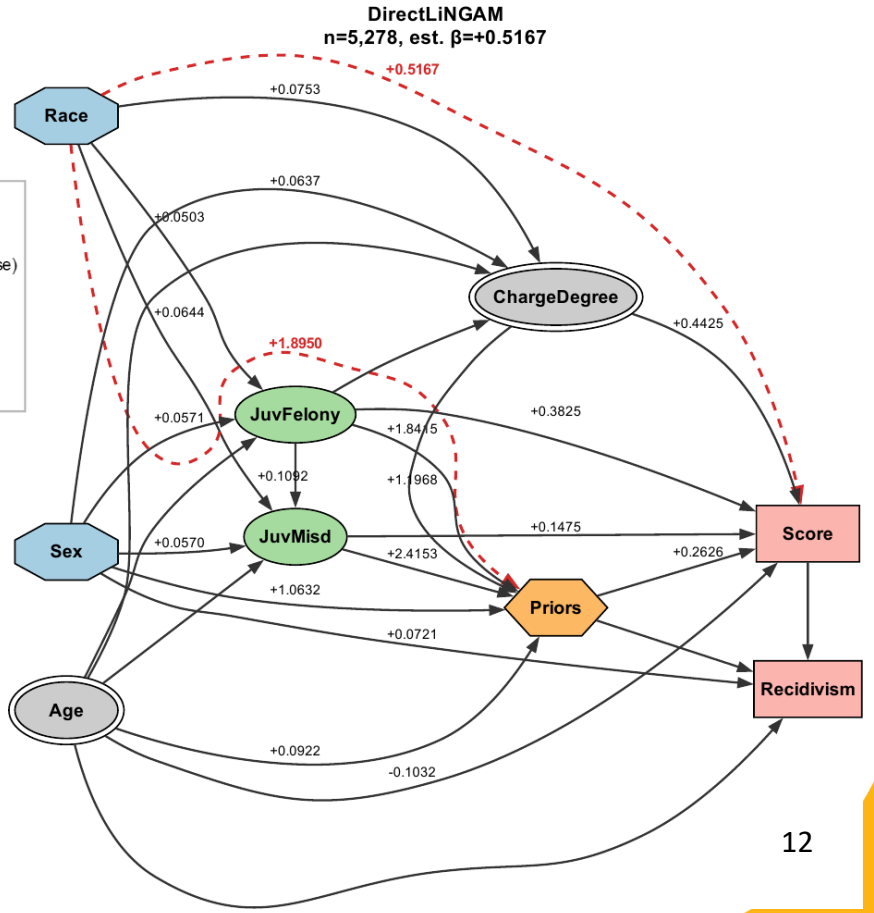


Step 3: Direct vs. Proxy Discrimination

COMPAS: Isolating the pathways

- Only LiNGAM recovers both pathways: direct Race → Score (+0.5167) and proxy Race → Priors (+1.8950)
- Unconstrained DirectLiNGAM rerun:
- β = +0.4834, so the edge is not an artifact of background knowledge
- GES and GRaSP orient Score → Race — impossible





Algorithm	Type	Race→Score	Race→Priors	Lat. conf.	β
PC	Constraint	No	Yes	False	–
FCI	Constraint+	No	No	True	–
GES	Score (BIC)	No	No	False	–
GRaSP	Permutation	No	No	False	–
ICA-LiNGAM	Func. Causal	Yes	Yes	False	+1.2105
DirectLiNGAM	**Func. Causal**	**Yes**	**Yes**	**False**	**+0.5167**

Step 4 Quantify: The Causal Effect

Total → Adjusted Direct Effect

Synthetic dataset

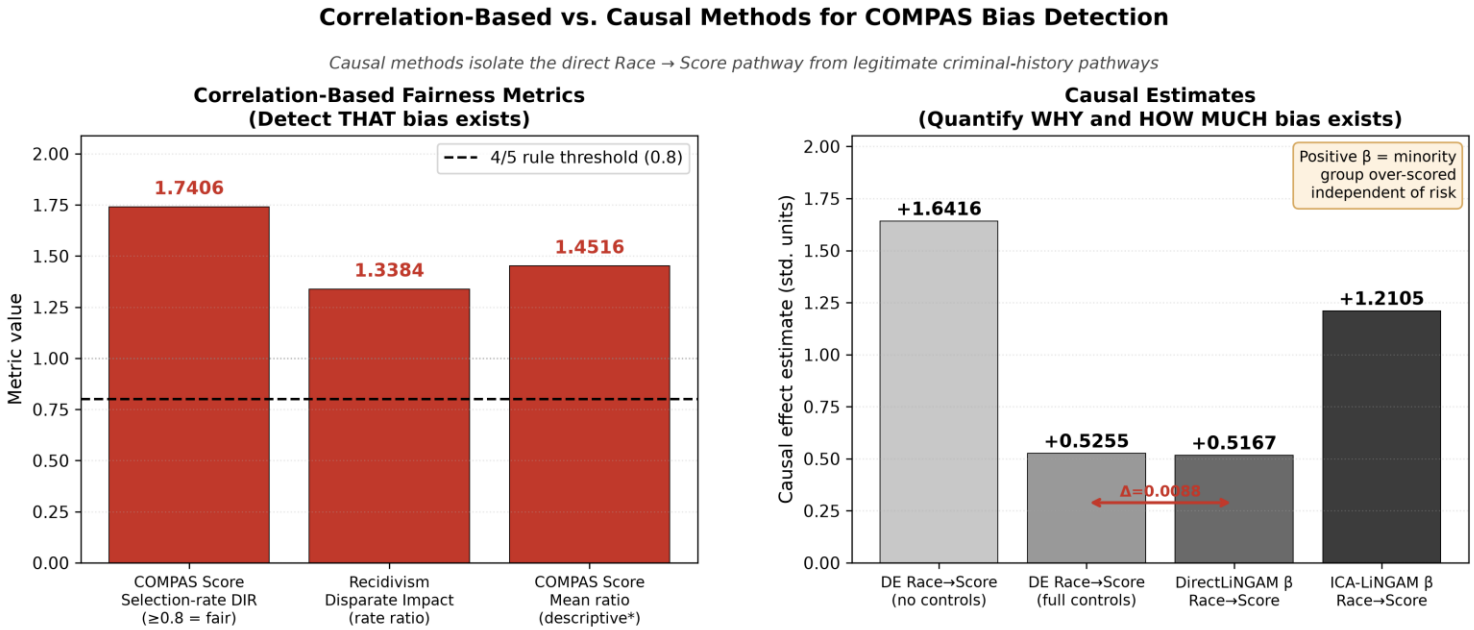
Stage	*Effect*	*SE*	*Controls (n)*	*Controls*
No controls	-0.2967	0.0297	0	None
+Income	-0.1691	0.0191	1	Income
+Income, CreditSc	-0.1075	0.0166	2	Income, Credit Score
+ZIP, Income, CreditSc	-0.1088	0.0181	3	ZIP, Income, Credit Score
+Full Controls	-0.1086	0.0182	5	Gender, Education, ZIP, Income, Credit Score

COMPAS dataset

Stage	*Effect*	*SE*	*Controls (n)*	*Controls*
No controls	+1.6416	0.0766	0	(none)
+Age + Charge	+1.1386	0.0725	2	Age, ChargeDegree
+Full criminal history	+0.5255	0.0638	5	Age, ChargeDegree, JuvFelony, JuvMisd, Priors

After full criminal-history controls, an adjusted direct effect of +0.5255 decile points (SE 0.0638, ≈ 0.18 SD) remains: African-American defendants scored higher than observationally identical Caucasian defendants. Synthetic check: DE −0.1086 vs. DirectLiNGAM −0.1074.

Step 5 - Validate: Bootstrapping and Sensitivity



DIR highlights discrimination bias but provides no mechanism to identify why or where.

- DirectLiNGAM +0.5167 vs. adjusted DE +0.5255 (Δ = 0.0088)
- Bootstrap 95% CI [+0.385, +0.637] excludes 0 (1,000 resamples)
- E-value ≈ 1.65 (1.52 at the CI lower bound) [27]

Conclusions & Future Work

- **Correlation shows that a disparity exists; causal methods show where it comes from** — and which intervention fits
- **Algorithm choice is not neutral:** LiNGAM, especially DirectLiNGAM, recovered the pathways other families missed
- **COMPAS:** A direct Race → Score effect (≈ +0.52 decile points) survives full controls; the proxy path through Priors needs upstream reform, not just dropping Race
- Arrows into immutable attributes are a cheap red flag for a BIC-driven search on algorithmic scores
- **Limitations:** COMPAS counts and binaries partly violate LiNGAM's linear, non-Gaussian assumptions — treat findings as hypotheses for domain and legal review
- **Future work:** Expanding the algorithms to lending, hiring, and healthcare data; kernel-based functional causal models to relax linearity

Thank you — Questions?

jpadela1@umbc.edu

Code, figures, tables: github.com/jpadela1/causal_bias_detection

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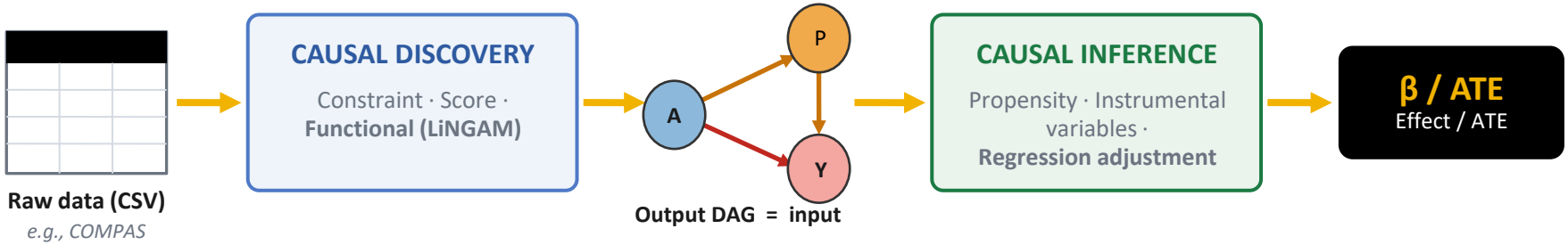
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Additional graphs and supplemental information

APPENDICES

Causal Methods



Example of using causal discovery and causal inference together to illustrate their use in our experiments. On the DAG, A is the input representation, P is the Proxy or Mediator, and Y is the Outcome.

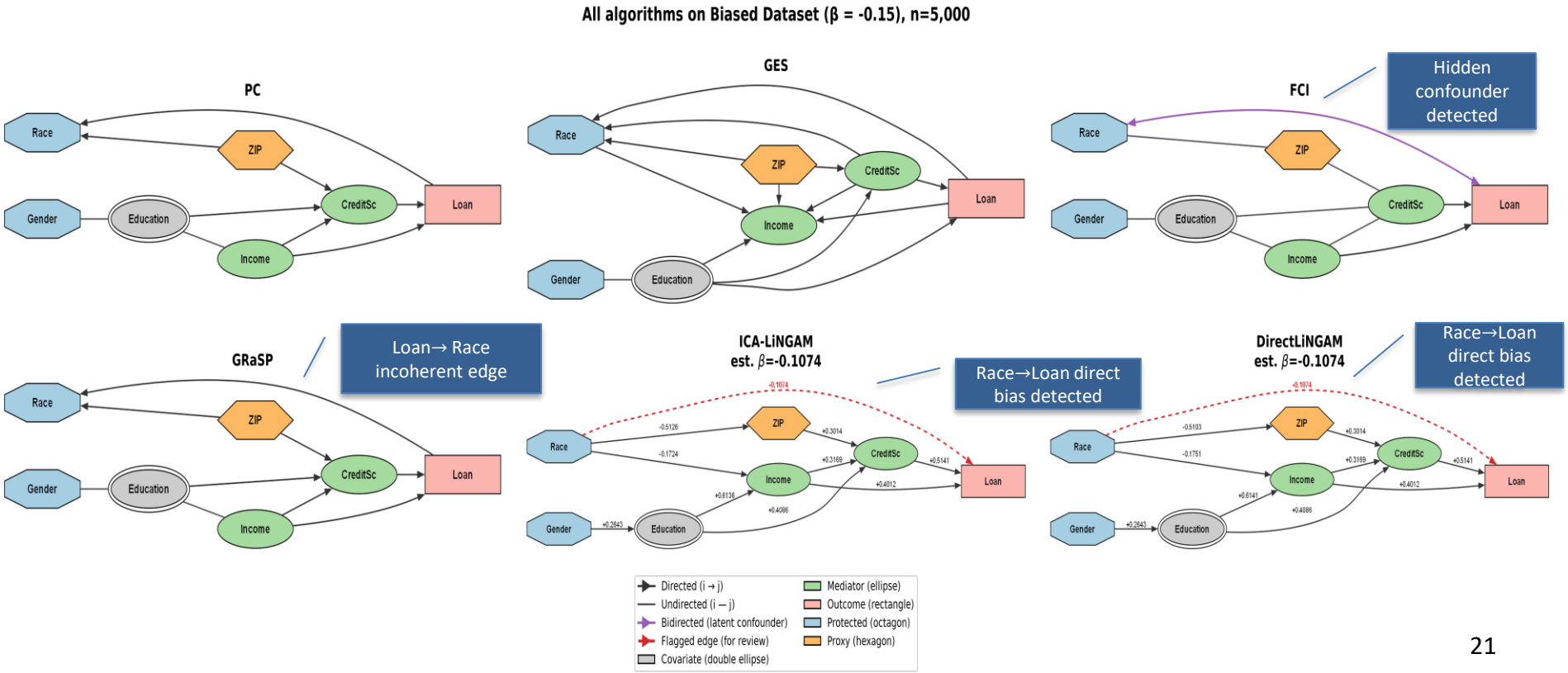
Causal discovery

- Input: Raw data (example, COMPAS 2-year recidivism records in a CSV file)
- Method: Constraint-, score-, or functional/structural equation-based models.
- Output: Directed Acyclic Graph (DAG) or Markov equivalence class (CPDAG) when the direction cannot be determined.

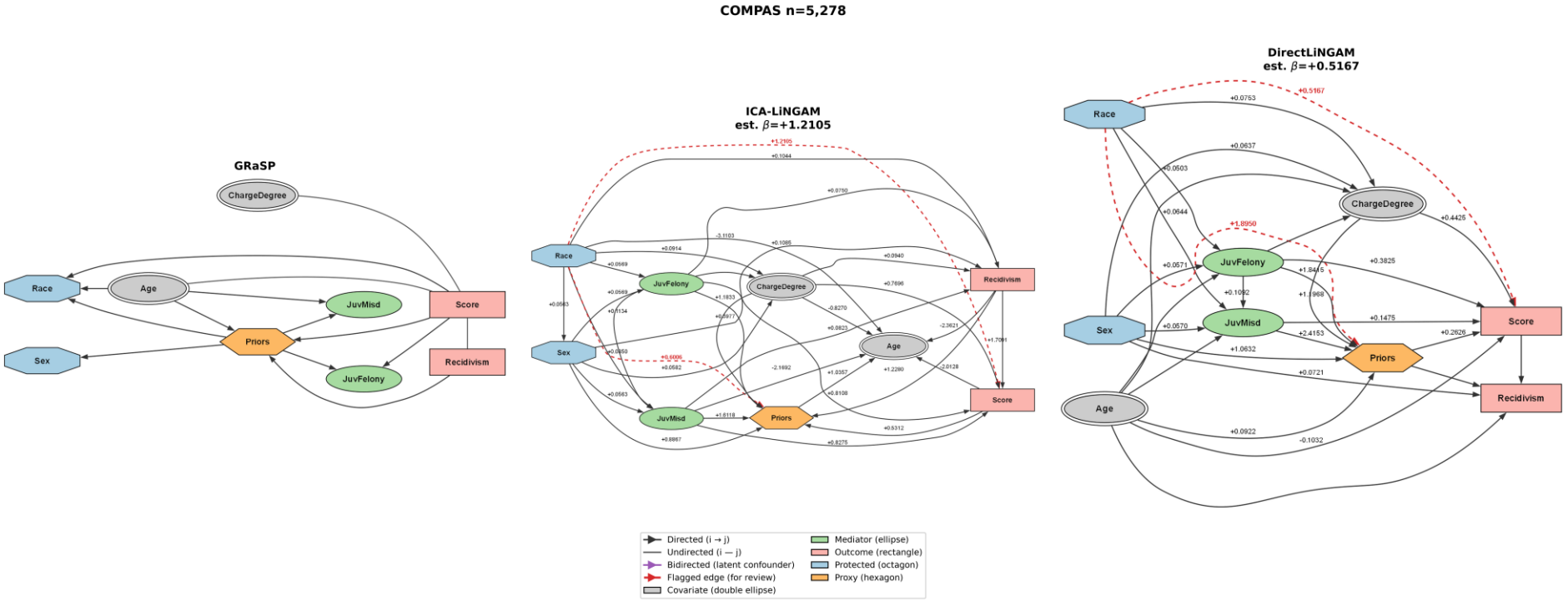
Causal inference

- Input: Domain knowledge, assumptions, or the output graph from causal discovery
- Method: Propensity scores, instrumental variables, and regression adjustments
- Output: An effect estimate – its size or direction; average treatment effect (ATE); or risk difference.

Causal Graphs Learned by the Six Algorithms *(Biased Dataset)*



Causal Graphs Learned by the Three of the Six Algorithms



Related Work

- Correlation-based fairness
  - Statistical parity – favorable outcome for an underprivileged group, 0.0 is equal parity. This is based on correlation and proportions.
  - Equalized odds/opportunity [6]
  - Disparate impact ratio (DIR [7]) – “4/5ths rule” – if the underprivileged group ratio is < 0.80 unfair practices and policies (hiring, lending) are likely.
    - Correlation-based fairness methods are comparisons based on ratios and proportions. However, when there are hidden confounders and proxy discriminators, it still results in biased outcomes.
    - Our method exposes causal pathways and hidden confounders. For example, in our lending dataset with planted bias, we expose the ZIP as a proxy discriminator.
- Causal Fairness
  - Counterfactual fairness (Kusner et al. [12]) – If one changed a protected attribute to the counterfactual, would the outcome change.
  - Predictive counterfactual fairness (Goethals et al. [13])
    - Both counterfactual fairness related works take a different approach than ours by applying counterfactual hypotheses to address fairness.
    - Both are limited in that getting rid of one attribute that addresses fairness, other proxy discriminators may remain. The proxies also create bias and unfair practices, such as reducing the likelihood of loan approval based on an applicant’s ZIP code.
    - These methods require increased oversight by domain experts who can guide the counterfactual selection (and can identify a proxy manually should one exist).

Related Work (continued)

- Causal Fairness (continued)
  - Survey of path-specific causal fairness methods (Su et al. [7]) – a review of limitations of correlations and advantages of causal-based methods. Our method uses causal discovery to detect bias and causal pathways that may expose discrimination through hidden confounders and proxy discriminators.
- COMPAS Study (Angwin et al. [3, 8])
  - We use this study to present the real-world dataset that has a high correlation with bias occurring in sentencing and judicial decisions.
  - We use comparable metrics to show the improved approach that causal science can provide.
- Causal discovery
  - Graphical model-based algorithms introduced in these two works (Shimizu et al.).
    - ICA-LiNGAM [15] – both LiNGAM variants identify the nodes, oriented edges, and shows the magnitude($|\beta|$) along the edges.
    - DirectLiNGAM [16] – we used both LiNGAM papers to inform our work and provide deeper analysis.
  - Causal learning (Cheng et al. [17])
    - We use causal discovery algorithms, but we experiment on a controlled bias scenario and then apply it to a real-world dataset with a reported bias
    - We wanted to see “if” causal discovery can detect bias, then which algorithms worked best, and to provide edge values to show the strengths or magnitude of the bias.
- Causal inference
  - Backdoor criterion to estimate an Average Treatment Effect (Pearl et al. [1, 2])
  - Cross-validation metrics and ATE results are used to further support or fail our methods.

Related Work: Correlation and Causal Fairness Methods - Our Novelty

- Prior fairness work shows disparity exists [6, 9], needs hand-built counterfactuals [12, 13], or remains theoretical [7, 26]. Our methods automatically discover the discriminatory pathways and suggest interventions at the dataset level, before prediction model training.

Existing AI Causal Fairness and Correlation Methods



Key Interpretations

- Constraint- (PC, FCI), score- (GES) and permutation-based (GRaSP) methods produce incoherent edges **if** they reliably detect the discrimination or proxy discrimination causal pathways at all.
  - FCI can detect confounders, but as we see in the COMPAS graph results, confounders may need further analysis.
  - GRaSP systematically fails due to mislabeling outcome variables as exogenous: causes, rather than something being caused (no parents).
- Both LiNGAM variants detected bias and proxy discrimination; however, DirectLiNGAM had greater accuracy in its oriented edges and in the sensitivity analysis ($n \geq 1{,}000$ and $\beta \geq 0.05$).
- Proxy vs. direct discrimination detection – the proxy discriminators achieve or influence the same outcome but through an indirect causal pathway.
  - Synthetic dataset proxy variable – ZIP, indirectly links Race to Loan Approval
  - COMPAS dataset proxy variable – Priors
- Both LiNGAM variants estimate the *standardized penalty,* the edge between variables, calculating the strength or magnitude of the influence of one variable upon another.
  - (-) for the synthetic dataset it serves to decrease loan approval likelihood
  - (+) for the COMPAS dataset the penalty increases the risk score for recidivism
- Sensitivity analysis assists in estimating the appropriate and consistent number of samplings required at various magnitudes or β values.
- Causal methods isolate the direct causal effect of Race→[COMPAS] Score pathway from the criminal history pathway.
- Both ICA- and DirectLiNGAM detect the Race→Score and the Race→Priors (proxy discriminators - discrimination that appears neutral but achieves the same outcomes as direct discrimination). We provide a method to detect both discriminatory pathways.
- With the increasing reliance on AI, automated decision support systems, and related emerging technologies, discrimination, bias identification, and mitigation strategies also increases the need for tools to combat unfair disadvantages, especially those pathways that are not identified through correlations alone.